EECS 16A Fall 2025 — Foundations of Signals, Systems, & Information Processing — Discussion 3B

1. Complex Exponentials & Euler's Formula!

(a) Represent $\cos(\omega t+\theta)$ and $\sin(\omega t+\theta)$ using complex exponentials. *Hint: Use Euler's formula:*

$$e^{i\phi}=\cos\phi+i\sin\phi$$

(b) **[Adapted from EE 120 Homework]** Prove De Moivre's Theorem, which states the following relationship for any $\theta\in\mathbb{R}, n\in\mathbb{Z}$:

$$(\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta$$

2. Orthogonality of Discrete-Time Complex Exponentials (Fourier Basis)

Let

$$x_k[n] = e^{i\frac{2\pi}{p}kn}, \quad n = 0, 1, \dots, p-1$$

for $k = 0, 1, \dots, p-1$. The complex inner product on $\mathbb{C}^p$ is

$$\langle a, b\rangle \ = \ \sum_{n=0}^{p-1} a[n]\,\overline{b[n]}.$$

(a) Show that

$$\langle x_k, x_m\rangle \ = \ 0 \quad \text{whenever } k \neq m.$$

(b) Explain why this orthogonality makes the set

$$\left\{x_k[n] = e^{i\frac{2\pi}{p}kn} \mid k \in [0, p-1]\right\}$$

a natural *basis* (the discrete Fourier basis) for $\mathbb{C}^p$.

